

Foundation (Jalapeno)

- Q1.** The equation $x^2 + 5x + 6 = 0$ has solutions of type
 (A) Two distinct real roots
 (B) One repeated real root
 (C) Two complex roots
 (D) No real roots
 (E) Cannot be determined
- Q2.** For the equation $x^2 - 3x - 2 = 0$, the discriminant is
 (A) 1
 (B) 13
 (C) 17
 (D) -9
 (E) 25
- Q3.** The quadratic equation $x^2 - 4x + 4 = 0$ represents a
 (A) Parabola with two x-intercepts
 (B) Parabola with one x-intercept
 (C) Parabola with no x-intercepts
 (D) Line
 (E) Circle
- Q4.** A basketball player scores x points in a game, where x satisfies the equation $x^2 - 12x + 36 = 0$. How many points did the player score?
 (A) 0 points
 (B) 6 points
 (C) 12 points
 (D) 18 points
 (E) 36 points
- Q5.** The equation $2x^2 + 5x - 3 = 0$ has solutions given by the quadratic formula as
- (A) $x = \frac{-5 \pm \sqrt{73}}{4}$
 (B) $x = \frac{-5 \pm \sqrt{49}}{4}$
 (C) $x = \frac{-5 \pm \sqrt{25}}{4}$
 (D) $x = \frac{-5 \pm \sqrt{37}}{4}$
 (E) $x = \frac{-5 \pm \sqrt{61}}{4}$
- Q6.** For the quadratic equation $x^2 + 2x + 5 = 0$, the discriminant is
 (A) -16
 (B) -4
 (C) 4
 (D) 16
 (E) 20
- Q7.** A cyclist travels x miles in a day, where x satisfies the equation $x^2 - 7x + 12 = 0$. How many miles did the cyclist travel?
 (A) 3 miles
 (B) 4 miles
 (C) 5 miles
 (D) 7 miles
 (E) 12 miles
- Q8.** The quadratic equation $x^2 - 9x + 20 = 0$ can be solved using the quadratic formula to get
 (A) $x = \frac{9 \pm \sqrt{19}}{2}$
 (B) $x = \frac{9 \pm \sqrt{49}}{2}$
 (C) $x = \frac{9 \pm \sqrt{39}}{2}$
 (D) $x = \frac{9 \pm \sqrt{29}}{2}$
 (E) $x = \frac{9 \pm \sqrt{41}}{2}$

Open Ended Questions (FRQ)

- Q9.** Solve the quadratic equation $x^2 + 4x + 4 = 0$ using the quadratic formula and determine the number and type of solutions.
- Q10.** A golfer hits a ball that travels x yards, where x satisfies the equation $x^2 - 10x + 16 = 0$. Solve for x and interpret the solutions in the context of the problem.

- Read each question carefully before answering.
- Use the quadratic formula to solve equations when necessary.
- Check your work to ensure accuracy.